

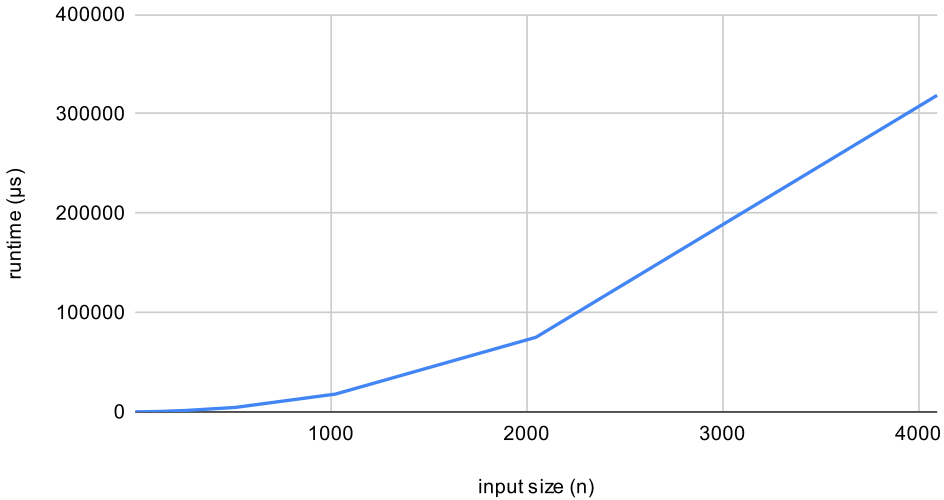
```
PS C:\Users\User\Algo-Abstraction\Assignment 1> ./cli.exe c 2 12
n = 2
4 microseconds
n = 4
4 microseconds
n = 8
8 microseconds
n = 16
17 microseconds
n = 32
37 microseconds
n = 64
124 microseconds
n = 128
362 microseconds
n = 256
1382 microseconds
n = 512
4386 microseconds
n = 1024
17832 microseconds
n = 2048
75000 microseconds
n = 4096
318418 microseconds
n = 2
INVALID: Out of range pair (2, 1).
144 microseconds
n = 4
INVALID: Out of range pair (4, 2).
155 microseconds
n = 8
INVALID: Out of range pair (8, 5).
175 microseconds
n = 16
INVALID: Out of range pair (2, 16).
125 microseconds
n = 32
INVALID: Out of range pair (32, 2).
122 microseconds
n = 64
INVALID: Out of range pair (64, 8).
164 microseconds
n = 128
INVALID: Out of range pair (128, 68).
290 microseconds
n = 256
INVALID: Out of range pair (256, 181).
330 microseconds
n = 512
INVALID: Out of range pair (512, 484).
920 microseconds
n = 1024
INVALID: Out of range pair (1024, 117).
3046 microseconds
n = 2048
INVALID: Out of range pair (2048, 481).
14495 microseconds
n = 4096
INVALID: Out of range pair (4096, 2357).
59744 microseconds
```

testMatching(2,10)		
n	runtime (µs)	
2	4	
4	4	
8	16	
16	32	
32	37	
64	124	
128	362	
256	1302	
512	4386	
1024	17832	
2048	75000	
4096	318418	

testVerify(2,10)		
n	runtime (µs)	
2	144	
4	155	
8	175	
16	125	
32	122	
64	164	
128	290	
256	330	
512	920	
1024	3046	
2048	14495	
4096	59744	

Both algorithms demonstrate polynomial time complexity

Matching Runtime



Verifying Runtime

